

Rajat Kumar

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Recent Ph.D. graduate in Chemistry with 6+ years of experience in molecular simulation, coarse-grained modeling, and statistical mechanics of complex molecular systems. My research connects molecular structure, phase behavior, transport, crystallization, vitrification, and interfacial organization in aqueous and polymer-containing systems, combining large-scale molecular dynamics, enhanced sampling, and reproducible Python-based analysis to extract physically interpretable molecular mechanisms.

EDUCATION

Ph.D., Chemistry

University of Utah, Salt Lake City, UT, USA

CGPA: 3.98/4.0

Aug 2020 – Apr 2026

Integrated BS–MS, Chemistry

Indian Institute of Science Education and Research (IISER) Kolkata, India

CGPA: 8.96/10.0

Aug 2015 – May 2020

RESEARCH EXPERIENCE

Graduate Research Assistant

Salt Lake City, UT

Prof. Valeria Molinero, University of Utah

Aug 2020 – Apr 2026

- Led large-scale molecular dynamics simulations of complex molecular systems to investigate phase behavior, slow dynamics, structural transformations, and metastability using coarse-grained and machine-learned coarse-grained models.
- Designed and implemented unbiased and enhanced-sampling simulations, including umbrella sampling and metadynamics, primarily in LAMMPS; developed reproducible Python workflows for structural, thermodynamic, transport, and interfacial analyses.
- Demonstrated competition between liquid–liquid transition and crystallization within a unified coarse-grained framework capable of capturing both phenomena (PNAS, 2024).
- Contributed to the first molecular-simulation study of medium-density amorphous ice, establishing its structural and thermodynamic signatures (PNAS, 2024).
- Developed a coarse-grained water–glycerol model to study crystallization–vitrification competition, transport anomalies, and structural heterogeneity in cryopreservation-relevant aqueous mixtures (Molecular Physics, 2024).
- Developed quantitative structural descriptors, including low- q correlation analysis, to identify physically meaningful observables and connect molecular simulations with experimentally measurable scattering signatures (J. Chem. Phys., 2026).
- Analyzed polymer–water/ice interfacial structure in a collaborative project on poly(vinyl alcohol) surfaces and ice recrystallization inhibition, connecting simulation-derived molecular mechanisms with experimental observables.
- Mentored two summer research interns (REU program) on coarse-grained modeling of aqueous salt solutions, guiding model development that remains stable across ambient and high-pressure conditions.
- Teaching Assistant, CHEM 3070 (Statistical Mechanics & Thermodynamics): led problem-solving sessions and supported 50+ students in ensemble theory, partition functions, and phase transitions.

Undergraduate Research Assistant (MS Thesis)*Kolkata**Kolkata, India**Prof. Pradeep Kumar Ghorai, IISER*

Aug 2015 – May 2020

- Investigated structure and dynamics of imidazolium-based ionic liquids across solvents spanning a wide range of polarity.
- Quantified the role of cation hydrophobicity in solvent organization using statistical-mechanical structural analysis of liquid mixtures.
- Developed Python-based analysis workflows to extract and interpret structural metrics for complex liquid systems.

MITACS Summer Research Intern*Calgary, Alberta, Canada**Prof. Peter Kusalik, University of Calgary*

May 2019 – Aug 2019

- Parameterized solvent-induced coarse-grained force fields for metal-organic frameworks (MOFs).
- Applied inverse Boltzmann inversion, inverse Monte Carlo, relative entropy minimization, and force matching to derive coarse-grained potentials.
- Implemented and validated coarse-grained models using GROMACS and MagiC to incorporate solvent effects into effective interaction potentials.

Indian Academy of Sciences Summer Research Fellow*JNCASR**Bengaluru, India**Prof. S. Balasubramanian,*

May 2018 – Jul 2018

- Performed a comparative evaluation of OPLS-2009IL and OPLS-VSIL force fields for imidazolium-based ionic liquids.
- Analyzed the impact of charge scaling and virtual-site representations on structural and dynamical properties.

SKILLS

- **Molecular Simulation & Modeling:** Classical molecular dynamics (all-atom and coarse-grained); enhanced sampling methods (umbrella sampling, metadynamics); multiscale modeling; long-timescale dynamics; phase-transition, crystallization, vitrification, and nucleation analysis.
- **Coarse-Graining & Force-Field Development:** Parameterization and validation of coarse-grained interaction models; inverse Boltzmann inversion, inverse Monte Carlo, relative entropy minimization, and force matching.
- **Complex Molecular Systems & Interfaces:** Aqueous mixtures, polymer-containing systems, polymer-water/ice interfaces, structural heterogeneity, transport properties, static structure factors, radial distribution functions, and structure-property relationships.
- **Computational Tools & High-Performance Computing:** LAMMPS, GROMACS, Python (NumPy, SciPy, Matplotlib), MDAnalysis, VMD, OVITO; reproducible data-analysis workflows; large-scale simulations on SLURM-based HPC clusters.
- **Machine Learning & Data-Driven Molecular Simulation:** Experience working with machine-learned coarse-grained molecular models; familiarity with PyTorch through formal training at the EMSL Summer School; foundational exposure to AI-assisted enhanced sampling, collective-variable learning, and statistical-mechanics-guided analysis of molecular simulation data.
- **Theoretical Foundations:** Statistical mechanics, thermodynamics, liquid-state theory, transport phenomena, phase transitions, and nonequilibrium processes.

- **Research Leadership & Collaboration:** Mentorship of undergraduate researchers; collaborative project development; scientific writing and presentation in interdisciplinary research settings.

PUBLICATIONS

- **R. Kumar**; I. de Almeida Ribeiro; D. Dhabal; V. Molinero. “*Rethinking the Evidence for a Liquid–Liquid Transition in Water: What Decompression Experiments Reveal.*” *The Journal of Chemical Physics* **164** (2), 024506 (2026).
- D. Dhabal*; **R. Kumar***; V. Molinero. “*Liquid–liquid transition and ice crystallization in a machine-learned coarse-grained water model.*” *Proceedings of the National Academy of Sciences* **121** (20), e2322853121 (2024).
- A. Zaragoza; **R. Kumar**; J. Martin Roca; P. Khatua; V. Molinero; F. Caupin; C. Valeriani. “*Anomalous behaviour of transport properties in a supercooled water–glycerol mixture.*” *Molecular Physics* **122** (21–22), e2413925 (2024).
- I. de Almeida Ribeiro; D. Dhabal; **R. Kumar**; S. Banik; S. K. R. Sankaranarayanan; V. Molinero. “*Medium-density amorphous ice unveils shear rate as a new dimension in water’s phase diagram.*” *Proceedings of the National Academy of Sciences* **121** (48), e2414444121 (2024).
- **R. Kumar**; A. Zaragoza; V. Molinero. “*Water polymorphism governs structural heterogeneity and crystallization in water–glycerol mixtures.*” to be submitted (2026).
- **R. Kumar**; D. Dhabal; I. de Almeida Ribeiro; V. Molinero. “*Competing Mechanisms of Liquid–Liquid Transition and Ice Crystallization in Supercooled Water.*” Manuscript in preparation.
- P. P. Taware; M.-J. Ferre; C. Crilly; E. Gubbins; Z. Brotherton; L. Rao; A. K. Kim; **R. Kumar**; B. Axtell; V. Molinero; B. Peters; K. Meister; M. Shultz; S. Han. “*Water structuring on poly(vinyl alcohol) surfaces underlies ice recrystallization inhibition activity.*” to be submitted (2026).

* Equal contribution (co-first authors).

INVITED TALKS, CONFERENCES & WORKSHOPS

- Invited Talk, Statistical Thermodynamics and Molecular Simulations (STMS) Virtual Seminar Series Apr. 22, 2026
- Gordon Research Conference: Water & Aqueous Solutions 2022, 2024
- Gordon Research Conference: Chemistry and Physics of Liquids 2025
- Berkeley Statistical Mechanics Meeting 2024
- EMSL Summer School (AI/ML, LLMs, and scientific agents) 2025
- Telluride School on Theoretical Chemistry 2021